DUAL-PLATFORM ELEVATION PLAN

Mobile Sales Closer & Web Studio Workbench: Comprehensive Flaw Analysis, UX Modernization & Implementation Roadmap

DOCUMENT CLASSIFICATION: Engineering Plan	**VERSION:** 2.0 (Post-Test Review)	**TARGET OS:** Flutter (Android/iOS) + Web (FastAPI/Three.js)	**DATE:** September 2026

1. Executive Summary & Problem Diagnosis

Following an end-to-end audit of our dual-platform architecture (Mobile Sales Closer + Web Studio Companion), we evaluated the commercial viability of the workflow against premier industry standards (Coohom, 2020 Design, Cabinet Vision). While the prototype successfully proves 15-minute consultation closing on a single straight wall, **14 critical architectural, spatial, and UX flaws prevent it from operating as an enterprise-grade commercial product**.

The core structural disconnect lies in the **unidirectional data flow and functional asymmetry**: the mobile app is touch-first but lacks project history, multi-wall support, and offline persistence; meanwhile, the Web Studio Workbench is strictly passive (view-only), preventing back-office estimators from modifying cabinet layouts, fixing on-site mistakes, or sending approved revisions back to field reps.

2. Comprehensive Flaw Audit & Target Solution Matrix

#	Area	Critical Flaw Identified	Operational & Closing Impact	Target Architectural Remedy
1	**Mobile**	**No Pipeline / Drafts Screen: Reps cannot view past quotes, reopen drafts, or search clients.**	Loss of sales continuity; cannot resume consultations across home visits.	**Consultations Hub: Mobile CRM screen with search, draft filters, and 1-tap reload.**
2	**Mobile**	**Blank Wall Blocker: `trace_screen.dart` disables 'Generate 3D' if obstacles list is empty.**	Reps cannot generate 3D for a plain wall without creating dummy windows/doors.	**Make obstacles optional; allow instantaneous 3D generation for plain walls.**
3	**Mobile**	**Pseudo Offline Storage: Claims to save locally on error, but writes no data to disk.**	If 4G/Wi-Fi drops on-site, signed quotes and finger signatures are permanently lost.	**Local Queue: `shared_preferences`/JSON draft outbox with auto-sync on reconnect.**
4	**3D Engine**	**1D Rail vs. Multi-Wall: Template generator offers L-Shape & Galley, but 3D engine only renders 1 straight wall.**	Reps cannot close L-shaped kitchens (>60% of modern residential homes).	**Multi-Wall 3D Rail: Extend Three.js to render corner walls (90°) with corner snapping.**
5	**Mobile**	**Blueprint Matrix Drift: `InteractiveViewer` pan/zoom does not transform Pin A & B coordinates.**	Pinch-zooming into blueprints misaligns calibration pins and corrupts scale ratio.	**Bind calibration pins to normalized image matrix space instead of screen coords.**
6	**Mobile**	**Mock WhatsApp & Sales ID: Sales ID hardcoded to 'demo_agent_01'; WhatsApp share is just a snackbar.**	Cannot track sales rep commissions; cannot instantly dispatch link to customer phone.	**Rep profile selector + native deep link (`whatsapp://send?phone=...`).**
7	**Web Studio**	**Passive / View-Only CAD: Estimators cannot move, snap, add, delete, or resize modules in Studio.**	Back-office designers cannot adjust field layouts without external software.	**Interactive CAD Editor: Drag, snap, delete, and insert cabinets directly in Studio.**
8	**Web Studio**	**No Revision Save UI: Backend has `PUT /api/projects/{id}`, but Web Studio has no UI to save edits.**	Design modifications cannot be persisted or pushed to factory CNC.	**Add 'Save Revisions', 'Override BOQ', and 'Approve Handover' controls.**
9	**Visuals**	**Primitive Box Graphics: Flat colored boxes without toe-kicks, benchtops, or hardware handles.**	Lacks emotional design appeal required to close \$15k–\$40k custom renovations.	**Continuous beveled countertops, 150mm recessed toe-kicks, and brushed handles.**
10	**API / Sync**	**Data Loss on Ingestion: `PlacedCabinet` schema omits `x_mm`, stripping module positions on POST.**	Future reloads and DXF exports collapse all cabinets to $X=0$.	**Add `x_mm`, `y_mm`, `z_mm`, and `wall_index` to Pydantic models.**
11	**Sync**	**Unidirectional Flow: Mobile pushes to backend, but cannot pull studio adjustments back.**	Field reps cannot see estimator corrections or approved manufacturer discounts.	**Bi-directional sync: Mobile can fetch updated project state via `GET /api/projects/{id}`.**

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12	**Client UX**	**No Shareable 3D Portal: Clients only receive a static PDF; no interactive web model for family review.**	Dampens closing velocity after the rep leaves the customer's living room.	**Client 3D Portal (`/view/{quote_id}`): Mobile web 360° spin, style toggle, and deposit link.**
13	**CPQ Engine**	**Missing Benchtops & Stone: CPQ prices cabinets and labour, but omits countertop materials.**	Benchtops account for 25%–35% of total project cost, creating massive quote inaccuracies.	**Add Benchtop Stone Calculator (Engineered Quartz, Granite, Laminate per linear metre).**
14	**Factory CAD**	**2D Wireframe DXF: DXF only contains basic lines without panel cutting lists or CNC nesting layers.**	CNC operators must manually re-draft parts before automated nesting.	**Layer-separated DXF with part IDs, carcass labels, and dimensional leaders.**

3. Deep Architectural Solutions & Technical Specifications

3.1 Mobile Consultation Hub & True Offline Architecture

To elevate the mobile app from a single-session demo into a true field sales tool, we replace the single-purpose intake screen with a dual-tab **Consultation Pipeline Hub**. Tab 1 houses the 3 room capture engines (Camera Wall Trace, 2D Blueprint Calibrate, and 10s Template Dial). Tab 2 displays a real-time list of all active consultations, allowing the rep to resume work, search by client name, or inspect status badges ('Priority Locked $250', 'Estimate Sent', 'Studio Revised').

- **Local Outbox Storage:** Quotes, placed cabinet coordinates, and base64 signatures are written to a persistent local JSON outbox using shared_preferences before network transmission. If connectivity fails, the app queues the submission and automatically flushes when Wi-Fi/4G is detected.
- **Zero-Obstacle Unblocker:** In trace_screen.dart, the check obstacles.isEmpty ? null : _generate3D() is removed. Empty walls immediately render in 3D, accelerating pitches for clean renovation surfaces.

3.2 Multi-Wall 3D Rail & Continuous Architectural Benchtop

Real kitchen sales are dominated by L-shape and U-shape spaces. We upgrade the Three.js scene (assets/3d/index.html) from a single 1D rail into a **Dual-Wall Coordinate System**:

- **L-Shape Geometry Generation:** When shape == 'lShape', the engine generates Wall 1 (Length W_1, aligned along X) and Wall 2 (Return Length W_2, rotated 90° and aligned along Z).
- **Corner Cabinet Snapping:** The corner_900 module automatically docks into the intersection $(0, 0, 0)$, anchoring the transition between Wall 1 and Wall 2.
- **Continuous Countertops & Kickboards:** Instead of each cabinet rendering its own independent worktop block, the engine generates a continuous stone slab spanning the entire placed base cabinet run, complete with 150mm recessed dark toe-kicks.

3.3 Interactive Web Studio Workbench 2.0 (Active CAD Editor)

The Web Studio companion (backend/static/studio/index.html) is upgraded from a passive viewer into a full **back-office CAD editing station**. Senior designers and estimators can:

- **Reposition & Snap Modules:** Click and drag cabinets along the wall rail using raycasting on desktop browsers, with live collision detection preventing cabinet overlaps.
- **Add / Delete / Resize Modules:** Add tall units, base drawers, or upper shelving directly from a sidebar palette, or delete misaligned items placed on-site.
- **Persistent Revision Control:** A 'Save Studio Revisions' action triggers PUT /api/projects/{id}, updating the financial BOQ, regenerating the factory DXF, and flagging the project as 'Studio Revised' for field reps.

3.4 Customer Shareable 3D Proposal Portal (`/view/{quote_id}`)

Closing in-person is powerful, but final multi-thousand dollar commitments often involve spouses, partners, or financial decision-makers who were not present. We introduce a public, lightweight web route /view/{quote_id}:

- **Mobile-First 360° Visualizer:** Loads directly in Safari/Chrome without app installation. Customers can spin the 3D kitchen, toggle finishes (Japandi, Nordic, Minimalist, Industrial), and inspect their exact cabinet inclusions.
- **Deposit & Slot Reservation:** Includes a direct call-to-action to lock in manufacturing queue priority ($250 AUD slot reservation via online payment gateway).

4. Five-Phase Execution Roadmap & Milestones

Phase	Focus Domain	Key Technical Deliverables	Validation Gate
Phase 1	**Mobile Polish & Core Data Integrity**	• Unblock empty wall 3D generation in `trace_screen.dart` • Mobile Consultation Pipeline Hub (view past quotes) • Real offline JSON outbox queue (`shared_preferences`) • Fix `PlacedCabinet` schema in `main.py` to preserve `x_mm`	**`flutter test`** **`pytest backend/test_studio.py`**
Phase 2	**Multi-Wall 3D & Spatial Realism**	• L-Shape multi-wall rail system in Three.js (`assets/3d/`) • Corner cabinet (900x900) junction snapping • Continuous beveled stone countertop generator • Benchtop stone calculator in `CPQEngine.dart`	**L-shape room visual test** **CPQ benchtop unit tests**

Phase 3	**Interactive Web Studio Workbench 2.0**	• Interactive cabinet drag, snap, and delete in desktop browser • Cabinet palette in Web Studio for back-office estimators • 'Save Revisions' calling `PUT /api/projects/{id}` • High-fidelity material textures (oak grain, matte slate)	**Playwright E2E recording of studio edits & DXF regeneration**
Phase 4	**Customer Shareable 3D Proposal Portal**	• Standalone customer route `/view/{quote_id}` • Native WhatsApp deep link from mobile app • Homeowner 360° interactive spin & style toggle • Online slot deposit payment trigger ($250 AUD)	**Mobile browser smoke test & WhatsApp payload verification**
Phase 5	**Precision Factory DXF & Production Nesting**	• AutoCAD Release 12 layered DXF with part IDs • Automatic dimensional leader lines on elevations • Itemized cutting schedule export (CSV/Excel)	**AutoCAD / LibreCAD import validation & CNC nesting check**

Architectural Approval & Sign-Off:

Prepared by: Antigravity Dual-Platform Engineering Team
Scope: Mobile Sales Closer, Web Studio Companion & Factory CNC Bridge

Status: Ready for Phased Execution

Recommended Starting Gate: **Phase 1 (Mobile Polish & Data Integrity)**
Target Delivery: Production-Grade Closing Velocity